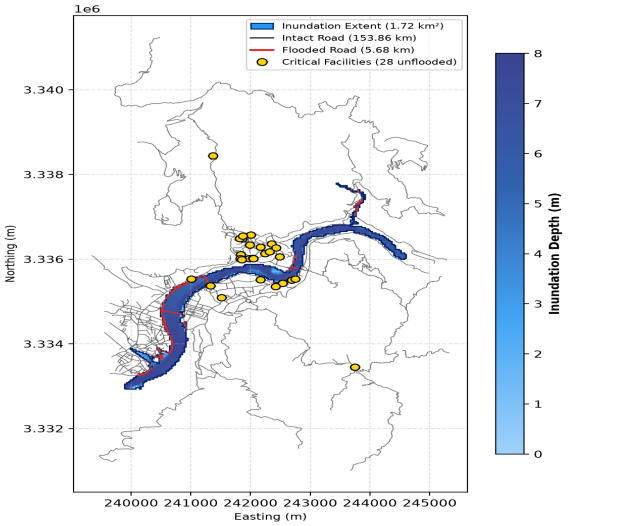
PravahX Dam Breach & Inundation Brief

Rishikesh Ganga Reach (Uttarakhand, India) | Tier-0 HAND Model

Run ID: tier0_ganga_accepted
Date: 2026-10-22 12:42 UTC
Status: ACCEPTED SCREENING

HYPOTHETICAL SCENARIO — TIER-0 SCREENING
Ganga Reach near Rishikesh (UTM 44N / EPSG:32644) | Peak Flow: 5,000 m³/s



DATA PROVENANCE & ENGINE:
- **DEM:** Copernicus GLO-30 (30 m STAC AWS)
- **CRS:** EPSG:32644 (UTM 44N) | **AOI:** 5×5 km
- **Hydraulic Forcing:** Peak Flow 5,000.0 m³/s, Manning n=0.035
- **Population:** WorldPop 2020 100m Constrained (BSGM IND)
- **Infrastructure:** OpenStreetMap (Snapshot: 2026-10-22, ODbL)

1. Hydraulic Screening Envelope

Inundation Area	**1.72 km² (171.9 ha)**
Peak Inflow Rate	**5,000.0 m³/s**
Maximum Depth	**7.73 m**
Mean Water Depth	**6.31 m**
Arrival Time	None (Static Tier-0)

2. Population & Asset Exposure

Exposed Population	**2,696 people**
• Depth 0.0–0.5 m [UNVERIFIED]	230 (8.5%)
• Depth 0.5–1.5 m [UNVERIFIED]	253 (9.4%)
• Depth > 1.5 m [UNVERIFIED]	2,213 (82.1%)
Road Network Flooded	**5.68 km of 159.5 km (3.6%)**
Flooded Buildings	**5 of 93 mapped**
Critical Facilities Inundated	**0 of 28 (depth = 0.00 m)**
Monetary Loss Valuation	**NOT RUN** (No verified JRC curve)

3. Network Evacuation Assessment

Vehicle Impassable Cutoff	Depth > 0.30 m [UNVERIFIED]
Severed Road Edges	**927** of 3638 edges
Reachable Cluster Origins	**1 (Route length: 108.8 m)**
Unreachable / Cut-off	**19 origins (Gorge road severed)**

MODEL LIMITATIONS & UNCERTAINTIES (MANDATORY DISCLAIMER):

1. ***Static Inundation Envelope:*** *Tier-0 HAND operates on static equilibrium and does not resolve hydrodynamic wave propagation, backwater effects, or wave arrival times.*
2. ***Synthetic Rating Curve:*** *Inundation depth is derived from normal-depth Manning hydraulics along a uniform reach without bathymetric cross-section survey calibration.*
3. ***DEM Water-Surface Artefact:*** *Copernicus GLO-30 reflects water surface elevation during satellite acquisition; dry-season channel bed conveyance is underestimated.*
4. ***Operational Scope:*** *Generated strictly for preliminary disaster risk screening and multi-agency table-top planning. NOT for structural engineering or design use.*